

Firebolt® 9 API Specification

Document status	DRAFT
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Note: this is a draft of the latest Firebolt spec

API version	Link	Summary
internal draft		

Interpretation

This page is an abstract, language-independent definition of the Firebolt Core API. The following terms are used:

	Term	Meaning
1	enum	an instance of a type that can have one of a pre-defined set of values. Typically represented by a string in JSON-RPC, but an integer in C++
2	list of	an ordered collection of values
3	list of... unordered	an unordered collection of values
4	one of	an instance of a primitive type, such as unsigned or string, that can have one of a pre-defined set of values. Not an enum
5	monotonic	a numeric value that is greater than the last instance of that attribute
6	generic error	an error that can be returned by any method call, documented in the Errors section
7	specific error	an error that can be returned by specific methods, documented along with the method

Key

Color	Meaning
Green	Approved - ready for development
Red	Not approved - not ready for development

Types

	Type	Definition
1	unsigned	32-bit unsigned integer
2	double	64-bit double-precision floating point number

Errors

	Class	Value	Name	Description	Examples	Open issues
1	Generic	-32601	Method not known	The method is not known to the version of Firebolt on this device	Device.foo	
2		-40300	Method not permitted	The app does not have permission to call this method		

3		-32602	Arguments invalid	The set of arguments is not valid		
4		-32100	Unspecified error	Any unexpected error	Memory allocation failure, transport failure	
5	Specific	-50100	Method not supported	A non-mandatory method that is not implemented on this device	SecureStorage.get on a non-Comcast device	
6		-xxxxx	App state invalid	The app is not in an appropriate lifecycle state to call this method		Sathish to assign the error value
7		-xxxxx	Speech synthesis engine error	No speech synthesis engine is available or the engine rejected the request		Sathish to assign the error value
8		-xxxxx	Wrong device class	The method is not appropriate for this class of device	App is incorrectly calling VideoOutput.cecState on a TV device	Sathish to assign the error value

Methods

	Module	Method	Parameters	Returns	Specific errors	API version	Deprecated API version	C++17	JS	Description
1	Accessibility	audioDescription onAudioDescriptionChanged	None	bool	None	8.0.0		✓	✓	Returns the audio description setting of the device
2	Accessibility	closedCaptionsSettings onClosedCaptionsSettingsChanged	None	- enabled - bool - preferredLanguages - list of strings [] (if not initialized) - list of one or more ISO 639-2 /B	None	8.0.0		✓	✓	Returns captions settings: enabled, and a list of zero or more languages in order of decreasing preference
3	Accessibility	highContrastUI onHighContrastUIChanged	None	bool	None	8.0.0		✓	✓	Returns the high contrast UI device setting
4	Accessibility	voiceGuidanceSettings onVoiceGuidanceSettingsChanged	None	- enabled - bool - rate - double 1 (normal rate) 0.1 to 10 inclusive - navigationHints - bool	None	8.0.0		✓	✓	Returns voice guidance settings: enabled, rate, and verbosity
5	Actions	start	- intent - json - handlerAppId - string - optional	None	None	9.0.0		✓	✓	Send an intent to the platform Argument is a JSON document
6	Actions	intent onIntent	None	- intentId - unsigned - monotonic - intent - json	None	9.0.0		✓	✓	Returns the intent that was most recently received from the platform, as a JSON document. Getter is typically called by an app after transitioning to the active lifecycle state
7	Advertising	advertisingId	None	- ifa - string - a UUID - ifa_type - string, one of "dpid" - device provided ID "sspid" - SSP provided ID "sessionid" - session /synthetic ID - limit - string, one of "0" "1"	None	8.0.0		✓	✓	Returns the IFA
8	Device	uid	None	string - a UUID	None	8.0.0		✓	✓	Returns a persistent unique UUID for the current app and device. The UUID is reset when the app or device is reset

9	Device	deviceClass	None	enum - ott - no tuner /demod, no integrated display - stb - with tuner /demod, no integrated display - tv - possibly tuner/demod, with integrated display	None	9.0.0		✓	✓	Returns the class of the device
10	Device	uptime	None	unsigned	None	9.0.0		✓	✗	Returns the number of seconds since most recent device boot, including any time spent during deep sleep
11	Device	timeInActiveState	None	unsigned	App state invalid	9.0.0		✓	✗	Returns the number of seconds since the device transitioned to the ON power state
12	Device	chipsetId	None	string - (see Chipset Id in Devices table)	None	9.0.0		✓	✗	Returns chipset ID as a printable string, e.g. "BCM72180"
13	Device	hdr onHdrChanged	None	- hdr10 - bool - hdr10Plus - bool - dolbyVision - bool - hlg - bool	None	8.0.0		✓	✓	Returns the HDR standards that are supported by the attached TV or the integral display - OTT/STB device: supported by the device and indicated by the EDID of the attached TV - TV: supported by the device
14	Device	dolbyAtmosExperienceAvailable onDolbyAtmosExperienceAvailableChanged	None	bool	None	9.0.0		✓	✓	Returns whether the user would get a Dolby Atmos experience if a Dolby Atmos track were to be played at this time
15	Device	manufacturer	None	string	None	9.0.0		✓	✓	Returns the manufacturer responsible for building the hardware
16	Device	brandName	None	string, either "" (if not initialized) 1 or more characters	None	9.0.0		✓	✓	Returns the brand name under which the device was marketed to consumers. Typically also shown on the TV bezel, device label or remote
17	Device	modelId	None	string	None	9.0.0		✓	✓	Returns the model identifier assigned to the device hardware. Typically also shown on the device label or UI
18	Device	osName	None	string	None	9.0.0		✓	✓	Returns the operating system name as defined by the operator
19	Device	osVersion	None	string	None	9.0.0		✓	✓	Returns the operating system version as defined by the operator
20	Device	firmware	None	string	None	9.0.0		✓	✓	Returns a string that identifies the firmware image of the device
21	Device	name onNameChanged	None	string	None	9.0.0		✓	✓	Returns the device friendly name. Used by network services (DIAL, Miracast, AirPlay) so that other devices can more easily identify this device during device discovery
22	Discovery	watched	- entityId - string - progress - double - optional - completed - bool - optional - watchedOn - ISO 8601 date and time in UTC - optional - agePolicy - string - optional	None	None	8.0.0		✓	✓	Notify the platform that content has been partially or completely watched progress: VOD: 0 to 0.99, live: number of seconds watchedOn: "YYYY-MM-DDThh:mm:ss.sssZ" agePolicy: app:adult, app:child, app:teen
23	Display	size	None	- width - unsigned - height - unsigned	None	9.0.0		✓	✗	Returns the physical dimensions of the connected or integral display, in centimeters Returns 0, 0 on a OTT/STB device when a display is not connected over HDMI Typical values 43" - 97, 56 55" - 123, 71 65" - 145, 83

24	Display	maxResolution	None	- width - unsigned - height - unsigned	None	9.0.0		✓	✗	Returns the physical/native resolution of the connected or integral display, in pixels Returns 0, 0 on a OTT/STB device when a display is not connected over HDMI Typical values - HD Ready - 1280, 720 - Full HD - 1920, 1080 - UHD - 3840, 2160
25	Display	edid	None	string - Base64	None	9.0.0		✓	✗	Returns the EDID (and extensions) of the connected or integral display, as a Base64 encoded string Returns an empty string, "", on a OTT/STB device when a display is not connected over HDMI
26	Display	colorimetry	None	list of enum, unordered - bt709 - bt2020	None	9.0.0		✗	✓	Returns an unordered list of colorimetry values supported by the attached TV or integral display. Returns an empty list if no TV is attached
27	Display	videoResolutions	None	list of enum, unordered 720p50 720p60 1080p50 1080p60 2160p50 2160p60	None	9.0.0		✗	✓	Returns an unordered list of HD video resolutions and frame rates supported by the attached TV or integral display. Returns an empty list if no TV is attached
28	Display	connected onConnectedChanged	None	bool	None	9.0.0		✓	✗	Returns whether the device is currently connected to an external or integrated display Most apps should instead use VideoOutput.hdcpc for display connectivity OTT/STB device: returns true if a TV is connected over HDMI (by hot plug detect pin 19), otherwise returns false TV device: returns true
29	Firebolt	apiVersion	None	- major - unsigned - minor - unsigned	None	9.0.0		✓	✓	Returns the version of the Firebolt API that is surfaced by the app gateway
30	Lifecycle2	close	type - enum - deactivate - unload - killReload - killReactivate	None	None	8.0.0		✓	✗	Request the platform to deactivate the app, and possibly take further action type - deactivate - app is deactivated (if active) - typically a user closing an app - unload - app is deactivated (if active) and terminated - typically an app has detected an error - killReload - app is killed, then reloaded to paused or suspended (by policy) - killReactivate - app is killed, reloaded, then activated window.close() maps to unload window.minimize() maps to deactivate Module is named Lifecycle in C++ Client Library
31	Lifecycle2	state	None	enum - initializing - active - paused - suspended - hibernated - terminating	None	8.0.0		✓	✗	Returns the current lifecycle state of the app. The first lifecycle state that can be observed by an app/runtime is the initializing state Apps/runtimes should ordinarily use state change notifications rather than this method
32	Lifecycle2	onStateChanged	List, of length one, of the lifecycle state change - oldState - enum - newState - enum	N/A	N/A	8.0.0		✓	✗	Subscribe to/unsubscribe from lifecycle state changes. The app/runtime remains in the initializing state until the subscribe call is made to the platform Notification of lifecycle state change, raised after the platform has transitioned the app/runtime to the new lifecycle state Note: the list always carries exactly one lifecycle state change Valid transitions - initializing to paused - initializing to suspended - paused to active - active to paused - paused to suspended - suspended to paused - suspended to hibernated - hibernated to suspended - any to terminating
33	Localization	country onCountryChanged	None	string, either "" (if not initialized) - ISO 3166-1 alpha-2 (see Country in Devices table)	None	8.0.0		✓	✓	Return the country, e.g. US, CA, GB, BE

34	Localization	preferredAudioLanguages onPreferredAudioLanguagesChanged	None	list of strings, either [] (if not initialized) - list of one or more ISO 639-2 /B (see Secondary Audio Language in Devices table)	None	8.0.0		✓	✓	A list of zero or more languages in order of decreasing preference. Typically two languages are present. A language may be repeated in the list
35	Localization	presentationLanguage onPresentationLanguageChanged	None	string, either "" (if not initialized) - BCP 47 (see Presentation Languages in Devices table)	None	8.0.0		✓	✓	The presentation language of the device, e.g. en-US
36	Localization	timeZone onTimeZoneChanged	None	string	None	9.0.0		✓	✗	Returns the current time zone in IANA format, e.g. America/New_York
37	Metrics	ready	None	None	None	8.0.0		✓	✓	Inform the platform that the app is minimally usable
38	Metrics	signIn	None	None	None	8.0.0		✓	✗	Log a sign in event
39	Metrics	signOut	None	None	None	8.0.0		✓	✗	Log a sign out event
40	Metrics	startContent	- entityId - string - optional - agePolicy - string - optional	None	None	8.0.0		✓	✓	Inform the platform that your user has started content
41	Metrics	stopContent	- entityId - string - optional - agePolicy - string - optional	None	None	8.0.0		✓	✓	Inform the platform that your user has stopped content
42	Metrics	page	- pageId - string - agePolicy - string - optional	None	None	8.0.0		✓	✓	Inform the platform that your user has navigated to a page or view
43	Metrics	error	- type - enum - network - media - restriction - entitlement - other - code - string - description - string - visible - bool - parameters - arg list - optional - agePolicy - string - optional	None	None	8.0.0		✓	✓	Inform the platform of an error that has occurred in your app
44	Metrics	mediaLoadStart	- entityId - string - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when setting the URL of a media asset to play, in order to infer load time
45	Metrics	mediaPlay	- entityId - string - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when media playback should start due to autoplay, user-initiated play, or unpausing

46	Metrics	mediaPlaying	- entityId - string - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when media playback actually starts due to autoplay, user-initiated play, unpausing, or recovering from a buffering interruption
47	Metrics	mediaPause	- entityId - string - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when media playback will pause due to an intentional pause operation
48	Metrics	mediaWaiting	- entityId - string - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when media playback will halt due to a network, buffer, or other unintentional constraint
49	Metrics	mediaSeeking	- entityId - string - target - double - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when a seek is initiated during media playback
50	Metrics	mediaSeeked	- entityId - string - position - double - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when a seek is completed during media playback
51	Metrics	mediaRateChanged	- entityId - string - rate - double - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when the playback rate of media is changed
52	Metrics	mediaRenditionChanged	- entityId - string - bitrate - unsigned - width - unsigned - height - unsigned - profile - string - optional - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when the playback rendition (e.g. bitrate, dimensions, profile, etc) is changed
53	Metrics	mediaEnded	- entityId - string - agePolicy - string - optional	None	None	8.0.0		✓	✓	Called when playback has stopped because the end of the media was reached
54	Metrics	event	- schema - uri - data - string - agePolicy - string - optional	None	None	8.0.0		✓	✓	Inform the platform of 1st party distributor metrics data parameter is a JSON document
55	Metrics	applInfo	build - string	None	None	8.0.0		✓	✓	Inform the platform about an app's build info
56	Network	connected onConnectedChanged	None	bool	None	8.0.0		✓	✓	Returns whether the device has a usable network connection

57	Network	interfaces	None	List of one or more objects, unordered • name - string • type - enum ◦ wired ◦ wifi • protocol - enum ◦ unknown ◦ 100base_tx ◦ 1000base_t ◦ 802_11ac ◦ 802_11ax ◦ 802_11be • enabled - bool • connected - bool • preferred - bool	None	9.0.0		✓	✗	Returns information about all physical network interfaces of the device name - e.g. wlan0 protocol - physical layer protocol supported by the interface, set to unknown by default connected - the link layer status preferred - true if the interface is used for network traffic
58	Network	interfaceStatus onInterfaceStatusChanged	interfaceName - string	• name - string • enabled - bool • connected - bool • preferred - bool	-xxxxx / Not found	9.0.0		✓	✗	Returns basic information about a network interface
59	Network	interfaceConfig onInterfaceConfigChanged	interfaceName - string	List of one or more objects, unordered • ipv4Address - string • ipv4DNSAddresses - list of zero or more strings • ipv6Addresses - list of zero or more strings • ipv6DNSAddresses - list of zero or more strings	-xxxxx / Not found	9.0.0		✓	✗	Returns the IP configuration of an interface
60	Network	interfaceStatistics	interfaceName - string	• interfaceStats - object - optional ◦ txPackets - unsigned null ◦ txError - unsigned null ◦ txDropped - unsigned null ◦ txFifoErrors - unsigned null ◦ txCarrierErrors - unsigned null ◦ rxPackets - unsigned null ◦ rxError - unsigned null ◦ rxDropped - unsigned null ◦ rxFifoErrors - unsigned null ◦ rxFrameErrors - unsigned null ◦ linkTxBitRate - unsigned null ◦ linkRxBitRate - unsigned null	-xxxxx / Not found	9.0.0		✓	✗	Returns statistics for a network interface interfaceStats - present when an interface is connected. Attributes are set to null if not available wirelessStats - present when an interface of type wifi is connected. Attributes are set to null if not available wirelessFrequency - MHz wirelessSignal - dBm, -96 (weakest) to 0 (strongest) inclusive wirelessQuality - current SNR in dB, 0 to 96 inclusive wirelessTxBitrate - Tx rate in Mbps wirelessRxBitrate - Rx rate in Mbps wirelessInactiveTime - time in milliseconds since last activity

				- wirelessStats - object - optional - wirelessFrequency - unsigned null - wirelessQuality - unsigned null - wirelessSignal - integer null - wirelessTxBitrate - unsigned null - wirelessRxBitrate - unsigned null - wirelessInactiveTime - unsigned null - wirelessRxBytes - unsigned null - wirelessRXPackets - unsigned null - wirelessRxDropped - unsigned null - wirelessTxBytes - unsigned null - wirelessTxPackets - unsigned null - wirelessTxRetries - unsigned null - wirelessTxFailed - unsigned null - wirelessExpectedThroughput - unsigned null						
61	Network	interfaceDetails onInterfaceDetailsChanged	interfaceName - string	- name - string - ssid - string - optional - bssid - string - optional	-xxxxx / Not found	9.0.0		✓	✗	Returns detailed information of a network interface Used to convey information which may be sensitive e.g personally identifiable information ssid - the SSID of the access point, provided for wifi interface type if connected bssid - the BSSID of the access point, provided for wifi interface type if connected
62	ParentalControl	pinControl	None	bool	-50100 / Not supported	???		✓	✓	Returns whether PIN blocking is enabled, or an error if this setting is not exposed to apps
63	ParentalControl	blockNotRatedContent	None	bool	-50100 / Not supported	???		✓	✓	Returns whether content that is not rated should be blocked, or an error if this setting is not exposed to apps
64	ParentalControl	viewingRestrictions	None	string	-50100 / Not supported	???		✓	✓	Returns a JSON document describing the ratings schemes configured for the device and any ratings that are blocked for that scheme, or an error if this setting is not exposed to apps
65	Presentation	focused onFocusedChanged	None	bool	None	8.0.0		✓	✗	Whether the app is in focus, i.e. receiving key presses. Provided for those apps/runtimes that cannot use Wayland
66	SpeechSynthesis	voices onVoicesChanged	None	List of zero or more, unordered - name - string - lang - string - BCP 47 - default - bool	None	9.0.0		✓	✗	Returns a list of available voices - name - name of the voice as a human-readable string - lang - the language of the voice in BCP 47 - default - true if the voice is the default for the language An empty list is returned if a speech synthesis engine is not available, or the engine rejected the request A particular voice can be repeated in the list, and a language can be repeated in the list At most one voice is the default for each language

67	SpeechSynthesis	speak	- text - string - lang - string - BCP 47 - optional - voice - string - optional - volume - double - optional - rate - double - optional - pitch - double - optional - pil - boolean - optional	unsigned	-xxxxx / Speech synthesis engine error	9.0.0		✓	✗	Requests the system to speak an utterance. Returns an utteranceId if the request is accepted, otherwise an error is returned. A successful new request is processed asynchronously: any utterance that is currently being spoken is interrupted and notified by onUtteranceEvent(interrupted), and events for the new request are raised using onUtteranceEvent - text - plain text or a well-formed SSML document - lang - the language of the text in BCP 47 - voice - as returned by the voices method - volume - the speaking volume, 0 (lowest) to 1 (highest) inclusive - rate - the speaking rate, 0.1 (slowest) to 10 (fastest) inclusive - pitch - the speaking pitch, 0 (lowest) to 2 (highest) inclusive - pil - true if the text contains personally identifiable information, indicating that functionality such as logging and telemetry should be restricted, false if not set For Voice Guidance, the app SHOULD either supply the text parameter as plain text and the lang parameter, or supply text as a well-formed SSML document. If text is plain text and lang is not supplied then the default speech synthesis language is used - this is not recommended usage
68	SpeechSynthesis	cancel	utteranceId - unsigned	None	-xxxxx / Speech synthesis engine error	9.0.0		✓	✗	Stop the utterance. If playing, audio is stopped. If synthesizing or paused, the utterance will not be played Raises onUtteranceEvent(interrupted) if the utterance was synthesizing, playing or paused Returns without error in all utterance states, or if the utterance has finished
69	SpeechSynthesis	pause	utteranceId - unsigned	None	-xxxxx / Speech synthesis engine error	9.0.0		✓	✗	Pause the utterance. If playing, audio is stopped. If the utterance was synthesizing, playback will be prevented Raises onUtteranceEvent(paused) if the utterance was synthesizing or playing Returns without error in all utterance states, or if the utterance has finished
70	SpeechSynthesis	resume	utteranceId - unsigned	None	-xxxxx / Speech synthesis engine error	9.0.0		✓	✗	Resume an utterance. If paused, audio is resumed. If synthesizing the utterance will be played. If playing no action is taken Raises onUtteranceEvent(resumed) if the utterance had previously been paused while playing Returns without error in all utterance states, or if the utterance has finished
71	SpeechSynthesis	onUtteranceEvent	- utteranceId - unsigned - event - enum - synthesisStarting - playbackStarting - paused - resumed - completed - interrupted - networkFailed - synthesisFailed - playbackFailed	N/A	N/A	9.0.0		✓	✗	Notifies events related to an utterance Progress events - synthesisStarting - text to audio conversion is being started - playbackStarting - audio playback is being started - paused - utterance has been paused while playing - resumed - an utterance that had previously been paused while playing has resumed playing - completed - the utterance has completed successfully - interrupted - the utterance has been stopped due to a call to either the speak or cancel methods Failure events - networkFailed - the system was unable to contact the voice synthesis backend - synthesisFailed - the system was unable to synthesize some or all of the utterance - playbackFailed - the system was unable to play some or all of the utterance

72	Stats	memoryUsage	None	• userMemoryUsed - unsigned • userMemoryLimit - unsigned • gpuMemoryUsed - unsigned • gpuMemoryLimit - unsigned	None	9.0.0		✓	✗	Returns information about container memory usage, in bytes
73	TextToSpeech	speak	text - string	• speechid - unsigned • TTS_Status - 0...3 • success - bool	None	8.0.0	9.0.0	✓	✗	Speak the utterance immediately. Any ongoing speech is interrupted Text argument is either plain text or a well-formed SSML document TTS_Status, not success attribute, to be used by caller to indicate success of call 0 OK, 1 Fail, 2 not enabled, 3 invalid configuration Raises onSpeechInterrupted if speaking is interrupted
74	TextToSpeech	pause	speechid - unsigned	• TTS_Status - 0...3 • success - bool	None	8.0.0	9.0.0	✓	✗	Pauses the utterance Raises onSpeechPause if ongoing speech is paused Does nothing if utterance is already paused
75	TextToSpeech	resume	speechid - unsigned	• TTS_Status - 0...3 • success - bool	None	8.0.0	9.0.0	✓	✗	Continue the paused utterance Raises onSpeechResume if paused speech is resumed Does nothing if the utterance is not paused
76	TextToSpeech	cancel	speechid - unsigned	• TTS_Status - 0...3 • success - bool	None	8.0.0	9.0.0	✓	✗	Stop speaking if utterance is currently being spoken Raises onSpeechInterrupted if speaking was interrupted
77	TextToSpeech	getSpeechState	speechid - unsigned	• speechstate - enum ◦ SPEECH_PENDING ◦ SPEECH_IN_PROGRESS ◦ SPEECH_PAUSED ◦ SPEECH_NOT_FOUND • TTS_Status - 0...3 • success - bool	None	8.0.0	9.0.0	✓	✗	Returns the state of the utterance
78	TextToSpeech	onWillSpeak	speechid - unsigned	N/A	N/A	8.0.0	9.0.0	✓	✗	Text to speech conversion is about to start
79	TextToSpeech	onSpeechStart	speechid - unsigned	N/A	N/A	8.0.0	9.0.0	✓	✗	Utterance is about to be spoken
80	TextToSpeech	onSpeechPause	speechid - unsigned	N/A	N/A	8.0.0	9.0.0	✓	✗	Ongoing speech was paused
81	TextToSpeech	onSpeechResume	speechid - unsigned	N/A	N/A	8.0.0	9.0.0	✓	✗	Paused speech was resumed
82	TextToSpeech	onSpeechComplete	speechid - unsigned	N/A	N/A	8.0.0	9.0.0	✓	✗	Speech completed successfully
83	TextToSpeech	onSpeechInterrupted	speechid - unsigned	N/A	N/A	8.0.0	9.0.0	✓	✗	Speech was stopped, due to another call to speak or cancel
84	TextToSpeech	onNetworkError	speechid - unsigned	N/A	N/A	8.0.0	9.0.0	✓	✗	Utterance failed due to network
85	TextToSpeech	onPlaybackError	speechid - unsigned	N/A	N/A	8.0.0	9.0.0	✓	✗	Utterance failed during playback
86	TextToSpeech	listVoices	language - string - BCP 47	• TTS_Status - 0...3 • voices - list of one or more strings	None	8.0.0	9.0.0	✓	✗	Returns the list of available voices as human-readable strings, e.g. "ava", "amelie", "angelica"
87	VideoOutput	resolution onResolutionChanged	None	• width - unsigned • height - unsigned One of • 720, 480 • 720, 576 • 1280, 720 • 1920, 1080 • 3840, 2160	None	9.0.0		✓	✓	Returns the width and height of the video signal on the video output, in pixels. Typically used by a streaming app to determine the highest resolution of video to select OTT/STB device: returns the video resolution over HDMI TV device: returns the resolution of the panel
88	VideoOutput	hdcp onHdcpChanged	None	• hdcp1.4 • hdcp2.2 • none • direct	None	9.0.0		✓	✓	Returns the current state of output protection on the video output OTT/STB device: returns the negotiated HDCP version on the video output, or none if an encrypted connection has not been made between the OTT/STB device and any attached TV TV device: returns direct

89	VideoOutput	cecState onCecState Changed	None	enum • active • inactive • unsupported	-xxxxx / Wrong device class	9.0.0		✓	✗	Returns whether the output of the OTT/STB device is actively being displayed on the TV OTT/STB device: returns the current state of the HDMI CEC connection • active - the TV is on and the OTT/STB device is the active source • inactive - the TV is off or the OTT/STB device is not the active source • unsupported - CEC is not enabled on the TV TV device: returns Wrong device class error
90	VideoOutput	refreshRate onRefreshR ateChanged	None	enum • 0 • 23.976 • 24 • 25 • 29.97 • 30 • 50 • 59.94 • 60	None	9.0.0		✓	✗	Returns the refresh rate of the video signal on the video output, in Hz OTT/STB device: returns the refresh rate, or 0 if no TV is attached TV device: returns the compositor refresh rate currently used to output content to the integrated display
91	VideoOutput	colorDepth	None	enum • 0 • 8 • 10 • 12	None	9.0.0		✓	✗	Returns the video color depth being used on the video output, in bits OTT/STB device: in addition, returns 0 when a display is not connected over HDMI TV device: does not return 0
92	VideoOutput	colorFormat	None	enum • ycbcr420 • ycbcr422 • ycbcr444 • rgb444 • none	None	9.0.0		✓	✗	Returns the current color format of the video signal on the active video output port or integrated display OTT/STB device: in addition, returns none when a display is not connected over HDMI TV device: does not return none
93	VideoOutput	colorimetry	None	enum • bt2020rgb • bt2020ycc • bt709 • oprgb • none	None	9.0.0		✓	✗	Returns the current colorimetry value of the video signal on the active video output port or integrated display OTT/STB device: in addition, returns none when a display is not connected over HDMI TV device: does not return none
94	VideoOutput	dynamicRan ge	None	enum • hdr10 • hdr10plus • dolbyVision • hlg • sdr • none	None	9.0.0		✓	✗	Returns the current dynamic range format of the video signal on the video output OTT/STB device: in addition, returns none when a display is not connected over HDMI TV device: does not return none
95	VideoOutput	quantization Range	None	enum • limited • full • none	None	9.0.0		✓	✗	Returns the quantization range used on the video output OTT/STB device: in addition, returns none when a display is not connected over HDMI TV device: does not return none
96	Window	addKeyInter cepts	• keys - list of one or more strings, unordered	None	None	9.0.0		✗	✓	Adds the supplied list of keys to the set of keys that are received by this app when it is in the Active lifecycle state but not in focus. The key is represented by the JS event.key attribute defined in the Firebolt Key Codes Specification
97	Window	removeKeyI ntercepts	• keys - list of one or more strings, unordered	None	None	9.0.0		✗	✓	Removes the supplied list of keys from the set of keys that are received by this app when it is in the Active lifecycle state but not in focus. The key is represented by the JS event.key attribute defined in the Firebolt Key Codes Specification